

PROJECT DEVELOPMENT PHASE

User Creation

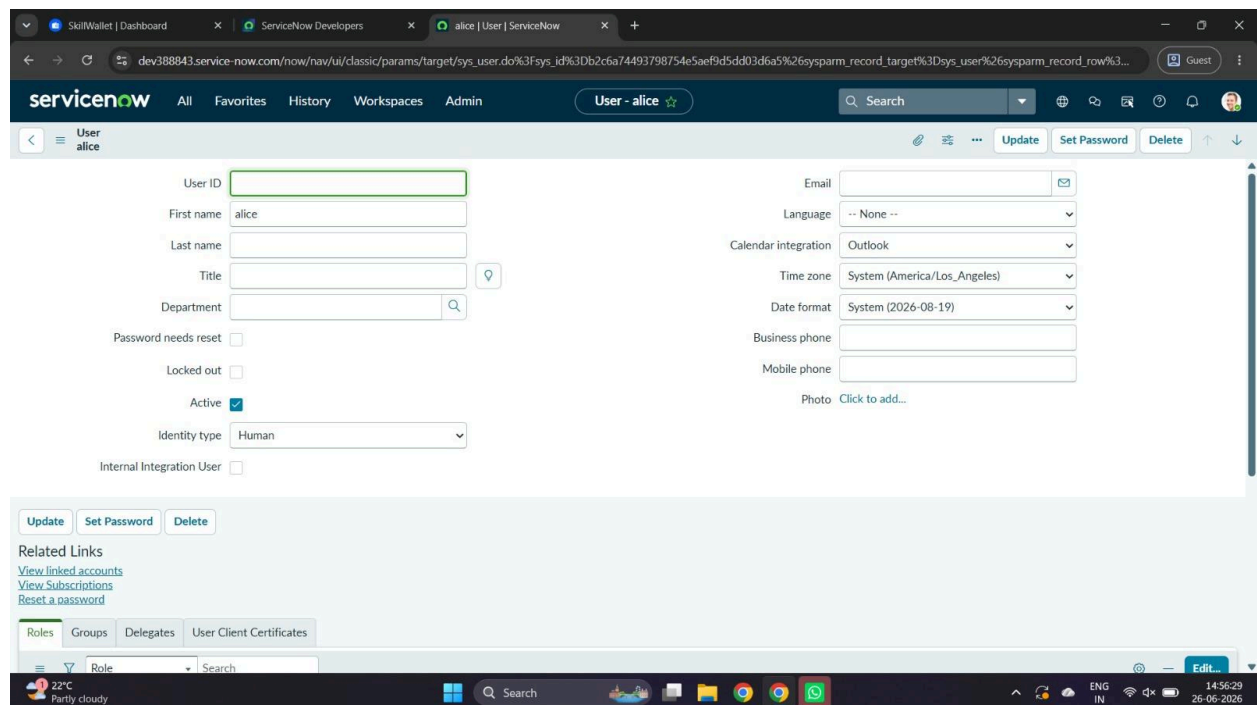
Objective

The objective of this phase is to create user accounts in ServiceNow that will participate in the workflow and access control process. These users represent different members of the organization and are assigned appropriate roles based on their responsibilities.

Explanation

In this project, two users named Alice and Bob were created in the ServiceNow instance. These users are used to test role assignments, group membership, workflow execution, and approval processes. Creating separate users helps simulate a real organizational environment where different employees perform different tasks.

User Creation – Alice



The screenshot shows the ServiceNow user creation interface for a user named 'alice'. The form is divided into two main sections: personal information and system settings. The personal information section includes fields for User ID, First name (alice), Last name, Title, and Department. The system settings section includes fields for Email, Language, Calendar integration, Time zone, Date format, Business phone, and Mobile phone. There are also checkboxes for 'Password needs reset', 'Locked out', 'Active' (checked), and 'Internal Integration User'. The 'Identity type' is set to 'Human'. At the bottom, there are tabs for 'Roles', 'Groups', 'Delegates', and 'User Client Certificates'. The 'Roles' tab is selected, showing a search bar and a list of roles. The ServiceNow logo and navigation menu are visible at the top, and the Windows taskbar is at the bottom.

User ID

First name

Last name

Title

Department

Password needs reset ☐

Locked out ☐

Active ☒

Identity type

Internal Integration User ☐

Email

Language

Calendar integration

Time zone

Date format

Business phone

Mobile phone

Photo

Update Set Password Delete

Related Links

[View linked accounts](#)

[View Subscriptions](#)

[Reset a password](#)

Roles Groups Delegates User Client Certificates

Role Search

The Alice user account was created successfully and is used for testing user management and access permissions within the application.

User Creation – Bob

The screenshot shows the ServiceNow user profile page for 'User - bob'. The form includes the following fields and options:

- User ID: Bob
- First name: bob
- Last name:
- Title:
- Department:
- Email:
- Language: -- None --
- Calendar integration: Outlook
- Time zone: System (America/Los Angeles)
- Date format: System (2026-08-19)
- Business phone:
- Mobile phone:
- Photo: Click to add...
- Password needs reset: ☐
- Locked out: ☐
- Active: ☒
- Identity type: Human
- Internal Integration User: ☐

Below the form are buttons for 'Update', 'Set Password', and 'Delete'. At the bottom, there are tabs for 'Roles (2)', 'Groups (2)', 'Delegates', and 'User Client Certificates'. A table with a 'Role' column is visible at the bottom of the page.

The Bob user account was created successfully and is used as the assigned user in workflow testing and approval scenarios.

Group Creation

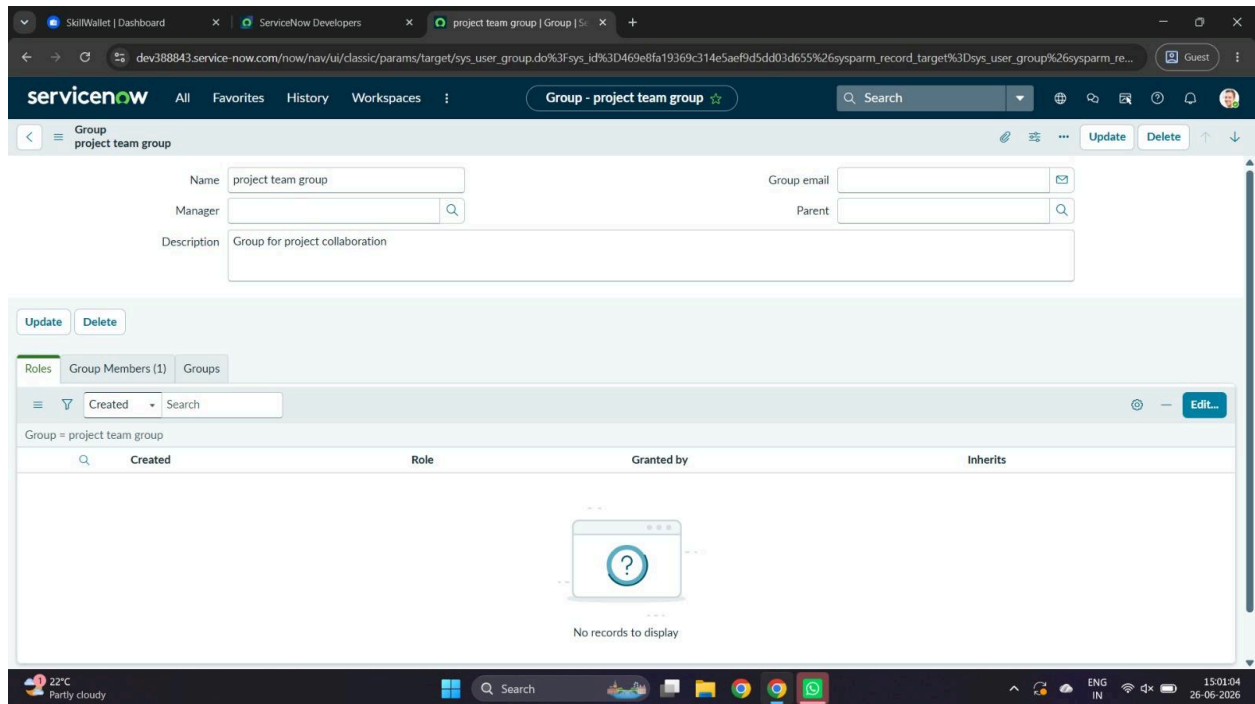
Objective

To organize users into a common group for simplified management and role assignment.

Explanation

A group named Project Team Group was created in ServiceNow. This group represents users working on the same project. By assigning users to a group, permissions can be managed more efficiently, reducing the need to assign roles individually.

Project Team Group Creation



The Project Team Group was created successfully, and users can now be assigned to this group for centralized access management.

Role Creation

Objective

To define roles that determine user responsibilities and system permissions.

Explanation

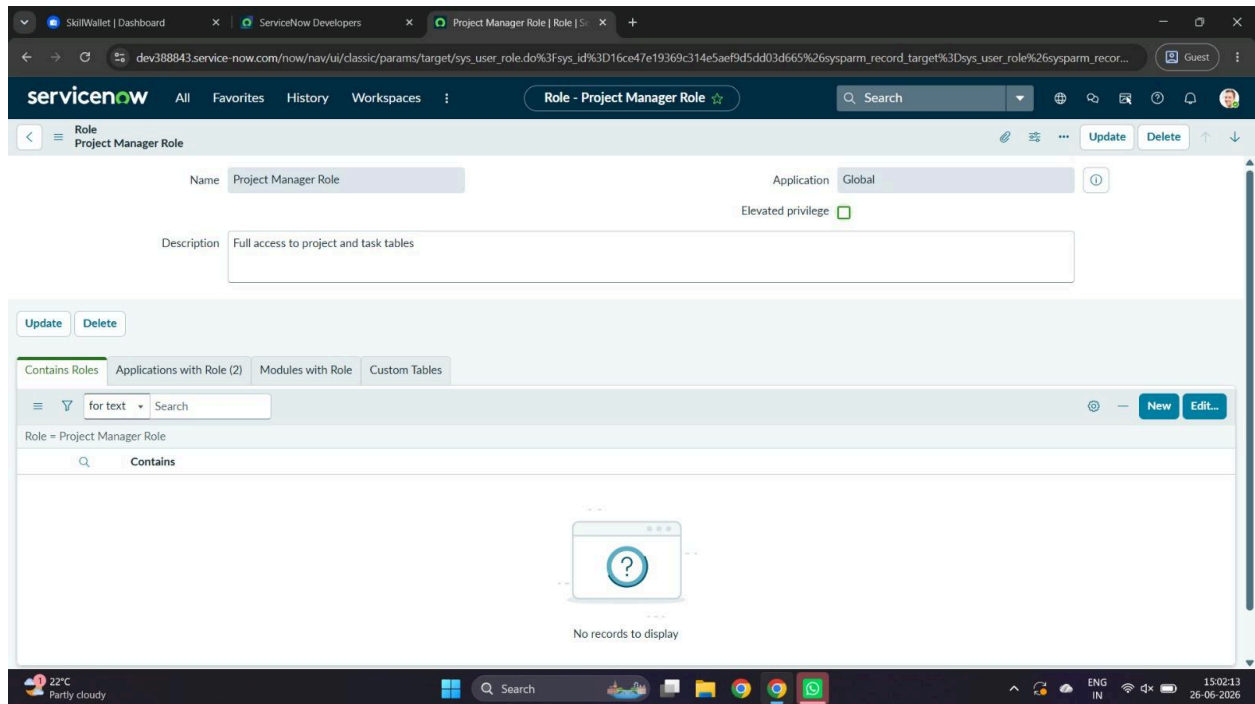
Two custom roles were created:

Project Manager Role

Team Member Role

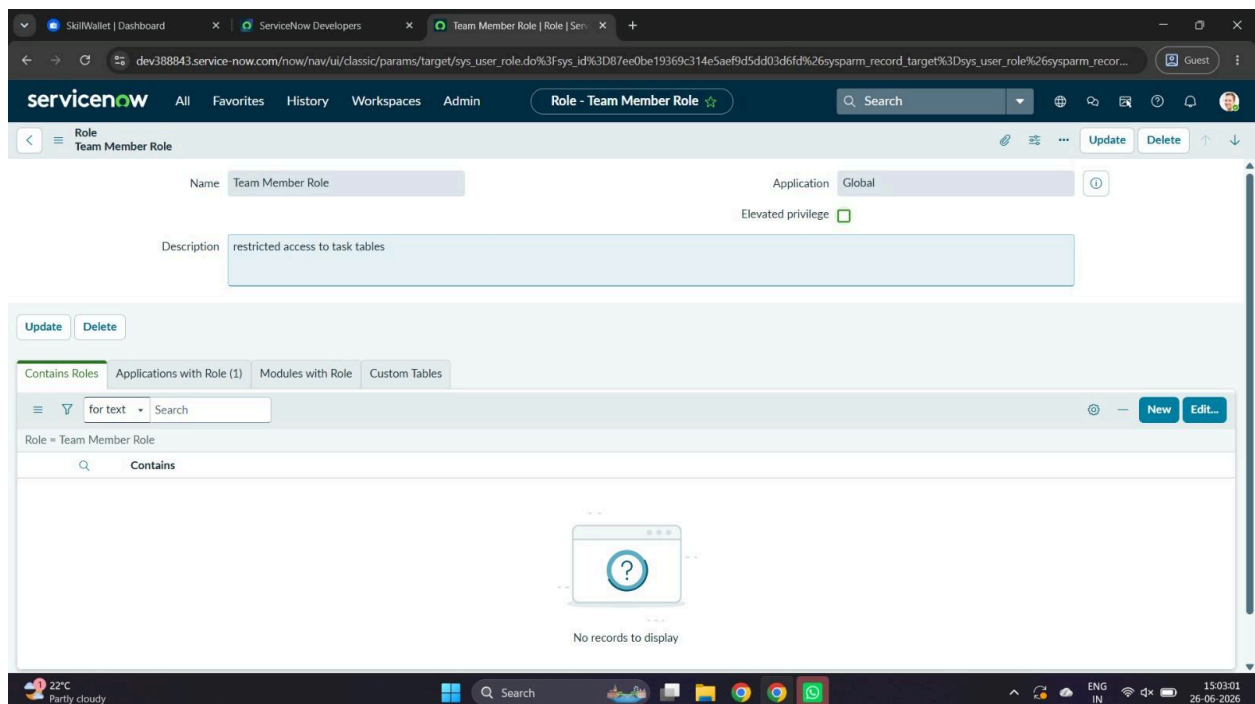
These roles help implement Role-Based Access Control (RBAC) by ensuring users receive only the permissions required to perform their tasks.

Project Manager Role



The Project Manager Role provides higher-level permissions required to manage project-related activities.

Team Member Role



The Team Member Role provides limited permissions required for team members to perform assigned tasks securely.

Assigning Roles to Users

Objective

To assign appropriate roles to users based on their responsibilities.

Explanation

After creating users and roles, the roles were assigned to the respective users. This ensures that users can access only the features and records permitted by their assigned role.

Role Assignment

Role assignment enables secure access control by providing users with permissions based on their organizational responsibilities.

Objective

To create custom tables for storing project and task information.

Explanation

Two custom tables were developed:

Project Table

Task Table 2

These tables store project details and task information required for workflow automation and approval management.

Project Table Creation

dev388843.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D1%26sys_is_list%3Dtrue%26sys_target%3Dsys_db_object%26sysparm_checked_items%3D%26syspar...

serviceNow All Favorites History Workspaces Admin Table - New Record Search

Table New record

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label project table

* Name u_project_table

Extends table

Application Global

Create module ☒

Create mobile module ☒

Add module to menu -- Create new --

New menu name project table

Remote Table ☐

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
X	project name	String				false
X	project manager	String				false
X	due date	Date				false
X	status	String				false
+	Insert a new row...					

22°C Mostly cloudy

Task Table 2 Creation

dev388843.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D80719b299369c314e5ae9d5dd03d625%26sysparm_record_target%3Dsys_db_object%26sysparm_rec...

serviceNow All Favorites History Workspaces Admin Table - Task Table 2 Search

Table Task Table 2

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Task Table 2

* Name u_task_table_2

Application Global

Remote Table ☐

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
	Created	Date/Time	(empty)	40		false
	Updated by	String	(empty)	40		false
	Updates	Integer	(empty)	40		false
X	comments	String	(empty)	40		false
	Updated	Date/Time	(empty)	40		false
X	description	String (Full UTF-8)	(empty)	50		false
X	task name	String (Full UTF-8)	(empty)	20		false
X	status	String	(empty)	40		false

1 to 11 of 11 New

T-storms In the morning

Assigning Roles to Project Table and Task Table

Objective

The objective of this phase is to associate the appropriate roles with the Project Table and Task Table 2 so that only authorized users can access and perform operations on these tables.

Explanation

After creating the custom tables, role-based permissions were configured. The Project Manager Role and Team Member Role were assigned to the respective tables to ensure secure access. This implementation follows the Role-Based Access Control (RBAC) model, where users can access only the data and functions permitted by their assigned roles.

Assigning roles to tables improves security by preventing unauthorized users from creating, modifying, or viewing sensitive project information.

Project Table with Assigned Role

Assigning Project Manager Role to Project Table

The screenshot shows the ServiceNow 'Application Menu - Project Table' configuration page. The page is titled 'Application Menu - Project Table' and includes a search bar and navigation links. The main configuration area contains the following fields and options:

- Title:** Project Table
- Application:** Global
- Active:** ☒
- Roles:** Project Manager Role
- Category:** Custom Applications
- Hint:** (empty text field)
- Description:** (empty text field)

At the bottom of the configuration area, there are 'Update' and 'Delete' buttons. Below the configuration area, there is a 'Modules' section with a search bar and a 'New' button.

The Project Table was configured with the Project Manager Role to ensure that only authorized users can manage project records.

Task Table 2 with Assigned Role

Assigning Team Member Role to Task Table 2

The screenshot shows the ServiceNow 'Application Menu - Task Table 2' configuration page. The page includes a header with the ServiceNow logo and navigation tabs. The main content area contains several sections: a title field set to 'Task Table 2', an application dropdown set to 'Global', an active checkbox checked, a roles section with a green checkmark and a list of roles including 'u_task_table_2_user', 'Project Manager Role', and 'Team Member Role', a category dropdown set to 'Custom Applications', and a hint/description section with empty text boxes. At the bottom, there are 'Update' and 'Delete' buttons, and a table header for 'Modules' with a search bar and a 'New' button.

The Task Table 2 was configured with the Team Member Role, allowing authorized team members to access and update task records securely.

Access Control Configuration

Objective

The objective of this phase is to implement Access Control Lists (ACLs) to protect sensitive information and restrict access based on user roles.

Explanation

Access Control Lists (ACLs) were created to provide field-level security for the Task Table 2. ACLs ensure that only users with the appropriate role can create, view, or modify specific fields. This prevents unauthorized access and strengthens the overall security of the application.

The following ACLs were configured:

Status

Comments

Assigned To

Description

Status ACL

The screenshot shows the 'Access Control - New Record' form in ServiceNow. The form is for creating a new record for the 'Status' field of 'Task Table 2 [u_task_table_2]'. The 'Type' is set to 'record' and the 'Operation' is 'create'. The 'Application' is 'Global'. The 'Active' checkbox is checked, and the 'Advanced' checkbox is unchecked. The 'Protection policy' is set to 'None --'. The 'Name' is 'Task Table 2 [u_task_table_2]' and the 'Status' field is selected. The 'Description' field is empty. A warning message at the top states: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' Below the form, the 'Conditions' section is expanded, showing a list of conditions. The 'Requires role' section is also expanded, showing a list of roles. The 'Team Member Role' is selected.

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record
* Operation: create
Admin overrides: ☒
Protection policy: -- None --
* Name: Task Table 2 [u_task_table_2]
status
Description:
Application: Global
Active: ☒
Advanced: ☐

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role

Team Member Role

The Status field ACL ensures that only authorized users can create or modify the task status.

Comments ACL

The screenshot shows the 'Access Control - New Record' form in ServiceNow. The form is for creating a new record for the 'Comments' field of 'Task Table 2 [u_task_table_2]'. The 'Type' is set to 'record' and the 'Operation' is 'create'. The 'Application' is 'Global'. The 'Active' checkbox is checked, and the 'Advanced' checkbox is unchecked. The 'Protection policy' is set to 'None --'. The 'Name' is 'Task Table 2 [u_task_table_2]' and the 'Comments' field is selected. The 'Description' field is empty. A warning message at the top states: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' Below the form, the 'Conditions' section is expanded, showing a list of conditions. The 'Requires role' section is also expanded, showing a list of roles. The 'Team Member Role' is selected.

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record
* Operation: create
Admin overrides: ☒
Protection policy: -- None --
* Name: Task Table 2 [u_task_table_2]
comments
Description:
Application: Global
Active: ☒
Advanced: ☐

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role

Team Member Role

Insert a new row...

The Comments field ACL restricts access, ensuring that only users with the required role can update task comments.

Assigned To ACL

Access Control - New Record

* Type: record

* Operation: create

Application: Global

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None --

* Name: Task Table 2 [u_task_table_2]

assigned to:

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
Team Member Role

Insert a new row...

The Assigned To ACL allows only authorized users to assign tasks to team members.

Description ACL

The screenshot shows the ServiceNow 'Access Control - New Record' form. The form is for creating a new record. The fields are as follows:

- * Type: record
- * Operation: create
- Application: Global
- Active: ☒
- Admin overrides: ☒
- Protection policy: -- None --
- * Name: Task Table 2 [u_task_table_2]
- Description: description

Below the form is a 'Conditions' section. It contains a blue box with the following text:

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

Below this is a 'Requires role' section. It contains a table with the following data:

Role
Team Member Role

The Description ACL protects project information by restricting unauthorized modifications.

Workflow Development using Flow Designer

Objective

The objective of this phase is to automate the project approval process using ServiceNow Flow Designer.

Explanation

A workflow was created using Flow Designer to automate task processing. The workflow begins automatically when the specified trigger conditions are met. It updates the task record, requests approval, and changes the task status based on the approval outcome.

Automation eliminates repetitive manual work, improves efficiency, and ensures that approvals follow a standardized process.

Workflow Trigger

The screenshot shows the 'task table' workflow trigger configuration in ServiceNow Workflow Studio. The trigger is set to 'Created' and is associated with 'Task Table 2 [u_task_table_2]'. The conditions are defined as follows:

- status is in progress
- comments is feedback
- assigned to is bob

The 'Data' panel on the right shows the flow variables for the trigger, including 'Task Table 2 Record', 'Task Table 2 Table', 'Run Start Time UTC', and 'Run Start Date/Time'. The 'ACTIONS' panel shows the 'Update Record' action.

The workflow is triggered automatically when the defined conditions are satisfied, initiating the approval process.

Update Record Action

The screenshot shows the 'task table' workflow update record action configuration in ServiceNow Workflow Studio. The action is set to 'Update Record' and is associated with 'Task Table 2 [u_task_table_2]'. The fields to be updated are defined as follows:

- status is completed

The 'Data' panel on the right shows the flow variables for the action, including 'Task Table 2 Record', 'Task Table 2 Table', 'Run Start Time UTC', and 'Run Start Date/Time'. The 'ACTIONS' panel shows the 'Update Record' action.

The Update Record action modifies the task information automatically before the approval request is initiated.

Ask for Approval Action

The screenshot displays the ServiceNow Workflow Studio interface for configuring a workflow. The main workspace shows the 'Ask For Approval' action selected. The 'Action Inputs' section includes fields for 'Record' (set to '1 - Updat...'), 'Table' (set to 'Task Table 2 [u_task_table_2]'), 'Approval Reason' (empty), 'Approval Field' (set to 'status'), and 'Journal Field' (set to 'Select a field'). Below these, the 'Rules' section shows two conditions: 'All users approve' and 'Anyone approves', both with 'alice X' as the user. The 'Due Date' is set to 'None'. The right-hand 'Data' pane shows the flow variables, including 'Trigger - Record Created' (Task Table 2 Record, Task Table 2 Table, Run Start Time UTC, Run Start Date/Time), '1 - Update Record' (Task Table 2 Record, Task Table 2 Table, Action Status), '2 - Ask For Approval' (Approval State, Action Status), and 'Error Handler'. The bottom status bar indicates 'Status: Draft' and 'Application: Global'.

The Ask for Approval action sends the task to the designated approver for review and approval.

Testing and Validation

Objective

The objective of this phase is to verify that the developed application functions correctly and meets the project requirements.

Explanation

Testing was conducted by creating sample task records and executing the workflow. Each component of the application, including user management, role assignment, ACLs, and workflow automation, was verified to ensure proper functionality.

The workflow executed successfully, approvals were processed correctly, and task records were updated automatically based on the approval results.

Update Record Testing

Screenshot of the ServiceNow 'Task Table 2s' interface. The browser address bar shows the URL: `dev388843.service-now.com/now/nav/ui/classic/params/target/u_task_table_2_list.do%3Fsysparm_userpref_module%3D45d197699369c314e5aef9d5dd03d6ad%26sysparm_clear_stack%3Dtrue`. The page title is 'Task Table 2s'. The table has columns: assigned to, comments, description, status, and task name. The table contains three rows of data:

assigned to	comments	description	status	task name
bob	feedback		Completed	
bob	feedback		requested	
bob	feedback		approved	

The bottom of the screenshot shows a Windows taskbar with the date and time: 15:41:59, 26-06-2026.

The Update Record action executed successfully, confirming that task details were updated automatically.

Approval Workflow Testing

Screenshot of the ServiceNow 'Approvals' interface. The browser address bar shows the URL: `dev223662.service-now.com/now/nav/ui/classic/params/target/sysapproval_approver_list.do%3Fsysparm_clear_stack%3Dtrue`. The page title is 'Approvals'. The table has columns: State, Approver, Comments, Approval for, and Created. The table contains one row of data:

State	Approver	Comments	Approval for	Created
Approved	alice		(empty)	2026-06-24 21:43:58

Below the table, it says: 'Number of rows removed from this list by Security constraints: 19'. The bottom of the screenshot shows a Windows taskbar with the date and time: 16:18:56, 26-06-2026.

The approval workflow was executed successfully, and the task status was updated according to the approval decision.